

Guide: Data Modelling Project (Power BI)

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What this file is

This Power BI model turns a messy set of tables into clean dimensions and facts, with a few measures, row-level security and one "protect the number" card. It was built in four phases: Prepare & Explore, Dimensions, Facts, Polish. This guide walks through the model for anyone opening the file.

The model at a glance

Facts (the events)

Table	What it holds	Connected to
fact_sales	One row per order line: quantity, price, cost, discount, line_total (200 lines, 80 orders)	dim_product, dim_customer, dim_order_flags, dim_geo (ship-to active, bill-to inactive)
fact_order_process	Order, ship, delivery, invoice and pay dates plus order_to_buy (days)	dim_customer
fact_inventory	Units per product per month	dim_product
fact_campaign_spend	Impressions, clicks and spend per campaign per day	dim_campaign
fact_promotion_coverage	Factless fact: which campaign promoted which product (no numbers, just the link)	dim_campaign, dim_product
fact_sales_targets	Monthly target_revenue (20 months)	nothing yet

Dimensions (the context)

Table	Notes
dim_customer	60 customers, with region (used by security)
dim_product	60 products, no duplicate product codes
dim_geo	20 cities, 5 regions; plays two roles (ship-to and bill-to)
dim_order_flags	The junk dimension: 14 combinations of channel, status and priority
dim_campaign	6 campaigns
dim_date	1 Jan 2025 to 31 Dec 2026 (730 days)

Support tables: security (user email to region) and _measure (a home for measures).

No two facts are connected directly. They all meet through shared dimensions, a core star-schema rule.

Measures

Measure	DAX
total_sales	SUM(fact_sales[line_total])
total_orders	DISTINCTCOUNT(fact_sales[order_id])
total_active_customers	DISTINCTCOUNT(fact_sales[customer_id])

Measure	DAX
base_total_customers	COUNT(dim_customer[customer_id])
avg_order_to_pay	AVERAGE(fact_order_process[order_to_buy])

Row-level security

One role, **Regional Access**, on dim_customer:

```
[region] = LOOKUPVALUE(security[region], security[user_email], USERPRINCIPALNAME())
```

The filter travels from dim_customer to fact_sales and fact_order_process. It does not touch the other facts: security is applied only where the requirements need it.

To test it: Modeling > View as > tick Regional Access > Other user > type nora.adel@arka.com. Every visual should show North America only. Try the other four emails in the security table too.

The number to protect

Total sales should read **526,643.91** (200 lines, 80 orders, 47 customers who bought). Keep it on a card and check it after every change to the model. Test customer 999 is not in the data.

Known limitations and next steps

These are planned improvements, not errors in the model.

1. **dim_date is not connected to any fact yet.** The order dates in fact_sales and fact_order_process carry a time of day (for example 13:50), so convert them to Date first, otherwise they will not match dim_date. Then relate fact_sales[order_date], fact_order_process[order_date], fact_campaign_spend[date], fact_inventory[month] and fact_sales_targets[period].
2. **fact_order_process has five date columns.** Keep order_date as the active relationship and make the others inactive, then use USERRELATIONSHIP in a measure. It is the same idea as bill-to city.
3. **dim_date only has year and month (as numbers).** Add month name, quarter and a sort column if the report needs them, and mark it as the date table.
4. **Auto date/time is still on.** Four hidden LocalDateTable tables are in the file. Once dim_date is connected, switch it off in Options > Current file > Data Load.
5. **Naming standards.** order_to_buy sits next to a measure called avg_order_to_pay, and invoices_date reads better as invoice_date. dim_order_flags has both channels and channel_code, so check that both earn their place.
6. **dim_customer holds contact details** (email, phone, street, contact_name). If no report uses them, the rule that every column must earn its place says to drop them.
7. **The card on Page 1** sums the line_total column directly. Point it at the total_sales measure instead.

How to explore the file

1. Open it in Power BI Desktop and go to **Model view** to see the diagram.
2. Click a relationship line to see its keys, cardinality and filter direction.
3. Open **Table view** to check each dimension's columns.
4. Use **Modeling > Manage roles** and **View as** for the security test.